Nazmul Huda Badhon

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Google Scholar || ResearchGate || LinkedIn || Website

Academic Qualification

Bachelor of Science in Computer Science and Engineering *(Jan'22 - Dec'25)*

Daffodil International University (DIU), Dhaka, Bangladesh

Thesis: Tabular data-based lightweight convolutional neural network

CGPA: 3.88/4.00 [Top 5% of cohort]

Exchange Semester in Computer Engineering *(Sep'23 - Dec'23)*

Dongseo University (DSU), Busan, South Korea

GPA: 3.94/4.50 [Received fully funded GKS-Global Korea Scholarship]

Research Interest

Deep Learning, Machine Learning, Computer Vision, Trustworthy AI

Research Summary

Computer Science and Engineering graduate with two years of research experience in deep learning and medical image analysis. Published eight peer-reviewed publications in top-tier journals. Experienced in neural network-based modeling and applied AI research. Seeking fully funded graduate research opportunities.

Research Experience

CSE AI Centre, Daffodil International University, Bangladesh *(Jan'26 - Present)*

Graduate Researcher

Advised by: Associate Professor Nazmun Nessa Moon and Professor Dr. Fernaz Narin Nur

- Develop and organize research datasets for applied artificial intelligence projects.
- Design and evaluate deep learning experiments for medical imaging and computer vision tasks.
- Prepare research manuscripts and support collaborative publication activities.

8th ABC Challenge, Kyushu Institute of Technology, Kitakyushu, Japan *(Dec'25 - Mar'26)*

Research Collaborator

- Analyzed real-world multimodal sensor data collected from a nursing care facility.
- Developed a Temporal Convolutional Network-based approach for sequence modeling.
- Classified indoor locations using BLE beacon signal data from healthcare.

Health Informatics Research Lab, Daffodil International University, Bangladesh *(Jan'24 - Dec'25)*

Student Researcher

- Developed deep learning models for medical image classification and segmentation tasks.
- Applied computer vision techniques to healthcare-related imaging datasets.
- Interpreted experimental results and contributed to research writing for publication.

Research Publications

Peer-Reviewed Journals:

1. **Nazmul Huda Badhon**, Sabit Ahmed Preanto, Abu Shahed Shah Md Nazmul Arefin, Kazi Jahid Hasan, and Md. Baharul Islam. Temporal convolutional network based indoor location recognition from ble beacon signals in nursing care facility. *International Journal of Activity and Behavior Computing* (2026). Japan Challenge Paper

2. **Nazmul Huda Badhon***, Imrus Salehin*, Md. Tomal Ahmed Sajib, Md. Sakibul Hassan Rifat, S M Noman, Nazmun Nessa Moon. TLCNN: tabular data-based lightweight convolutional neural network for electricity energy demand prediction. *Global Energy Interconnection* (2025), Elsevier. Q1

3. Md. Tomal Ahmed Sajib, **Nazmul Huda Badhon**, Imrus Salehin, Md. Sakibul Hassan Rifat, Faysal Ahmmed, Nazmun Nessa Moon. A comparative deep learning methodology for plant insect image classification: assessment of CNN architectures and augmentation techniques. 103680, *MethodsX* (2025), Elsevier. Q2

4. Imrus Salehin, Md. Tomal Ahmed Sajib, **Nazmul Huda Badhon**, Md. Sakibul Hassan Rifat, Nazrul Amin, Nazmun Nessa Moon. Systematic Literature Review of LLM-Large Language Model in Medical: Digital Health, Technology and Applications. 7.9: e70365, *Engineering Reports* (2025), Wiley. Q1

5. Imrus Salehin, **Nazmul Huda Badhon**, Md Tomal Ahmed Sajib, Mohd Saifuzzaman, and Nazmun Nessa Moon. IQ-UltraRecon: Demodulated IQ ultrasound dataset of human hand and arm tissue for deep learning-based reconstruction. *Data in Brief* (2025), 112001, Elsevier. Q1

6. Imrus Salehin, **Nazmul Huda Badhon**, Md Tomal Ahmed Sajib, and Nazmun Nessa Moon. DeepUS-ReconSeg: A multi-angle paired B-mode Ultrasound dataset for medical imaging reconstruction and segmentation. *Data in Brief* (2025), 112083, Elsevier. Q1

7. Imrus Salehin, Mahbubur Rahman Khan, Ummya Habiba, **Nazmul Huda Badhon**, Nazmun Nessa Moon. BAU-Insectv2: An Agricultural Plant Insect Dataset for Deep Learning and Biomedical Image Analysis. *Data in Brief* (2024), 110083, Elsevier. Q1

Book Chapter:

8. Imrus Salehin, **Nazmul Huda Badhon**, Md. Tomal Ahmed Sajib, Md. Sakibul Hassan Rifat, Nazmun Nessa Moon. Predicting Peak Energy Demand Using Machine Learning Techniques for Efficient Electricity Supply Management. *In Engineering Applications of AI for Demand Forecasting*, Vol. 1, p. 13. Taylor & Francis Group, 2026.

Manuscripts Under Review:

9. A. H. M. Saiful Islam*, **Nazmul Huda Badhon***, Fernaz Narin Nur*, Md. Tomal Ahmed Sajib, Nazmun Nessa Moon. Bridging Neuroscience and Education: A Scoping Review of EEG-Based Cognitive and Emotional States in Educational Settings. Manuscript no: IJEDRO-D-26-00680 (*International Journal of Educational Research Open*, Elsevier) Q1

** Joint co-first authors*

Projects

Selected Research Projects

1. Breast Cancer Classification using Hybrid Segmentation Deep Learning
- Developed a hybrid U-Net and CNN model.
- Evaluated model performance using accuracy and ROC-based metrics.

2. Lung and Colon Cancer Histopathology Classification
- Applied preprocessing and transfer learning techniques to histopathology images.
- Evaluated classification performance using accuracy and F1-score.

3. Vision Transformer–Based Medical Image Classification
- Implemented a ViT-based classification pipeline for medical image analysis.
- Achieved 87% test accuracy.

Additional Projects

4. Lost and Found Web Application
- Built a full-stack web application for reporting and retrieving lost items.
- Used HTML, CSS, JavaScript, PHP, and MySQL with image upload functionality.

Technical Skills

Programming : *Python, Java, C*
DL Techniques : *CNN, U-Net, Vision Transformers, Transfer Learning*
Frameworks : *PyTorch, TensorFlow, Keras, Scikit-learn*
Data Analysis : *NumPy, Pandas, Matplotlib, Seaborn*
Research Tools : *GitHub, Google Colab, LaTeX*

Language Proficiency

- Bangla: Native
- English: Professional working proficiency (Academic research, writing and presentation).

Academic Service

- Peer Reviewer, Digital Health, SAGE Publishing, 2025

Honors & Awards

- Research Award for Scholarly Publications, *Division of Research, DIU, 2024-2025.*
- Best Performer Award, *Computer Vision and Medical Image Analysis Bootcamp, HIRL, DIU, 2025.*
- Global Korea Scholarship awardee, *Government of the Republic of Korea, 2023.*
- Merit-based Scholarship, *Daffodil International University, 2022-2025.*
- Government General Board Scholarship, *Education Board Bangladesh, 2015*

Leadership and Extracurricular Activities

- **Team Leader** *2023*
  International Camp, Dongseo University, South Korea

- **Volunteer Intern** *2024*
  International Affairs, Daffodil International University

- **Founding General Secretary** *(Mar'24 - July'24)*
  DIU Research Society

References

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  Department of Computer Science and Engineering
  Daffodil International University, Dhaka, Bangladesh
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